

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	15 February 2025
Team ID	AGRI-ANALYTICS-01
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As a data analyst, I can collect crop production data from government sources (data.gov.in, Ministry of Agriculture, etc.)	3	High	Data Engineer
Sprint-1	Data Preprocessing	USN-2	As a data analyst, I can clean and preprocess the collected data to handle missing values and inconsistencies	2	High	Data Engineer
Sprint-2	Exploratory Data Analysis	USN-3	As a data analyst, I can perform EDA to understand crop production trends across states and years	3	High	Data Analyst
Sprint-2	Visualization & Dashboard	USN-4	As a user, I can view interactive visualizations and dashboards of crop production by state, crop type, and year	5	High	Data Analyst, Frontend Dev
Sprint-3	Insights & Reporting	USN-5	As a user, I can generate insightful reports on top-producing crops, states, and year-on-year growth	3	Medium	Data Analyst
Sprint-3	Predictive Analysis	USN-6	As a user, I can use predictive models to forecast crop production for future years	5	Medium	ML Engineer
Sprint-3	User Authentication	USN-7	As a user, I can register and log in to access personalized dashboards	2	Low	Backend Dev
Sprint-4	Deployment	USN-8	As a user, I can access the application deployed on the cloud	2	Medium	DevOps Engineer
Sprint-4	Documentation	USN-9	As a developer, I can document the project for future reference and maintenance	1	Low	Technical Writer

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	10 Days	20 Feb 2025	01 Mar 2025	18	01 Mar 2025
Sprint-2	20	10 Days	02 Mar 2025	11 Mar 2025	20	11 Mar 2025
Sprint-3	20	10 Days	12 Mar 2025	21 Mar 2025	19	21 Mar 2025
Sprint-4	20	10 Days	22 Mar 2025	31 Mar 2025	-	-

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{Sprint Duration}}{\text{Velocity}} = \frac{20}{10} = 2$$

Average Velocity (per day) = 2 Story Points